

Bioinformatics

Project Kick-Off

Category	Lung adenocarcinoma (LUAD)	Lung squamous cell carcinoma (LSCC)
Description	A lung cancer arising from glandular or secretory epithelial cells in the lung	A lung cancer originating from squamous epithelial cells that line the airways.
Typical location	Peripheral lung (near alveoli)	Central lung (near bronchi)
Smoking association	Modreate	Strong
Occurrence in non-somkers	Common	Rare
Common genetic alterations	EGFR, KRAS, ALK, BRAF, MET	TP53, PIK3CA, FGFR1, CDKN2A
Frequency	~40-50% (most common)	~20-30%
Targeted therapy	Widely available	Limited
Datasets		
Training data	Protein and RNA expression in tumor: 88 patients Protein and RNA expression in normal adjacent tumor (NAT): 88 patients Overall survival table: 88 patients	Protein and RNA expression in tumor: 74 patients Protein and RNA expression in normal adjacent tumor (NAT): 74 patients Overall survival table: 74 patients
Test data	Protein and RNA expression in tumor: 9 patients Protein and RNA expression in normal adjacent tumor (NAT): 9 patients Overall survival table: 9 patients	Protein and RNA expression in tumor: 8 patients Protein and RNA expression in normal adjacent tumor (NAT): 8 patients Overall survival table: 8 patients

Three-Layer Project

Tumor vs. Normal Classification (20 points)

- Input: Protein and/or RNA expression (tumor and normal)
- Output: Tumor or Normal

LUAD vs. LSCC Classification (20 points)

- Input: Tumor protein and/or RNA expression (tumor only)
- Output: LUAD or LSCC

Survival Prediction (10 points)

- Input: Tumor protein and/or RNA expression (tumor only)
- Output: Survival or Death

Test

- Please share your Github repository link on eCampus.
- Please make sure to prepare clear and well-organized documentation/manuals.
- The projects will be tested and the scoreboard will be updated every Friday late evening.